POF Workbench Practical Guide

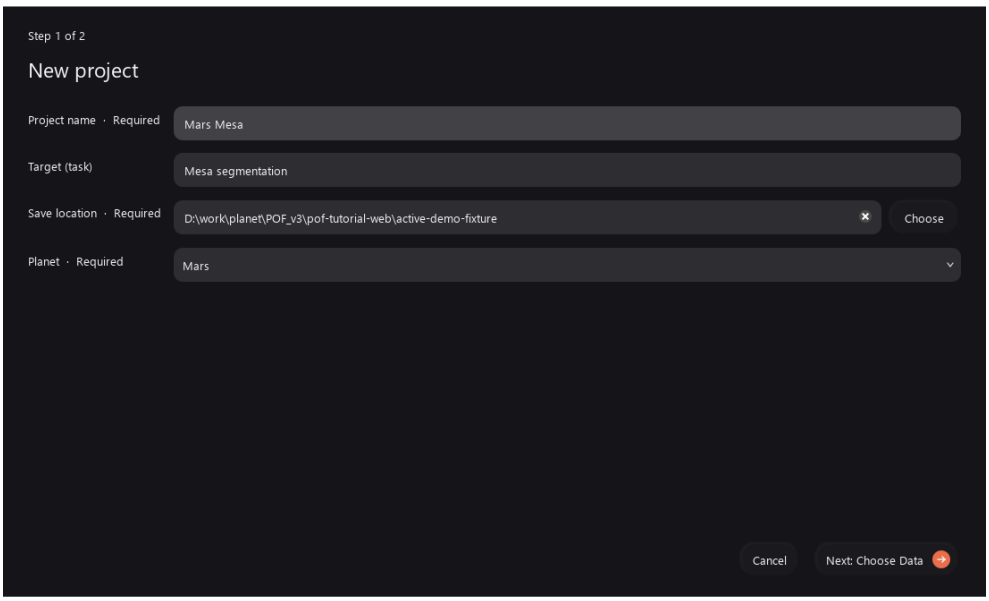
Annotation, training, active learning and batch inference. Follow the English interface controls shown in the illustrations.

1. Projects and data

A project groups research work. Each task keeps its own image sources, labels, models and active-learning rounds.

1. Choose **New project** and enter the name, location, target and planet. Use **Switch project** to open an existing project.
2. Choose **Enter task** in the task list. Opening a project does not automatically select a task.
3. Select **Active learning**, **Training** or **Inference**, add the required data and run **Check data**.

Training images and labels must have matching names and pixel dimensions. Unmatched files appear in the check details; valid pairs can still be used. Label normalization creates separate copies and preserves the originals.



Project and task entry. Select a task before preparing data.

2. Annotation and label editing

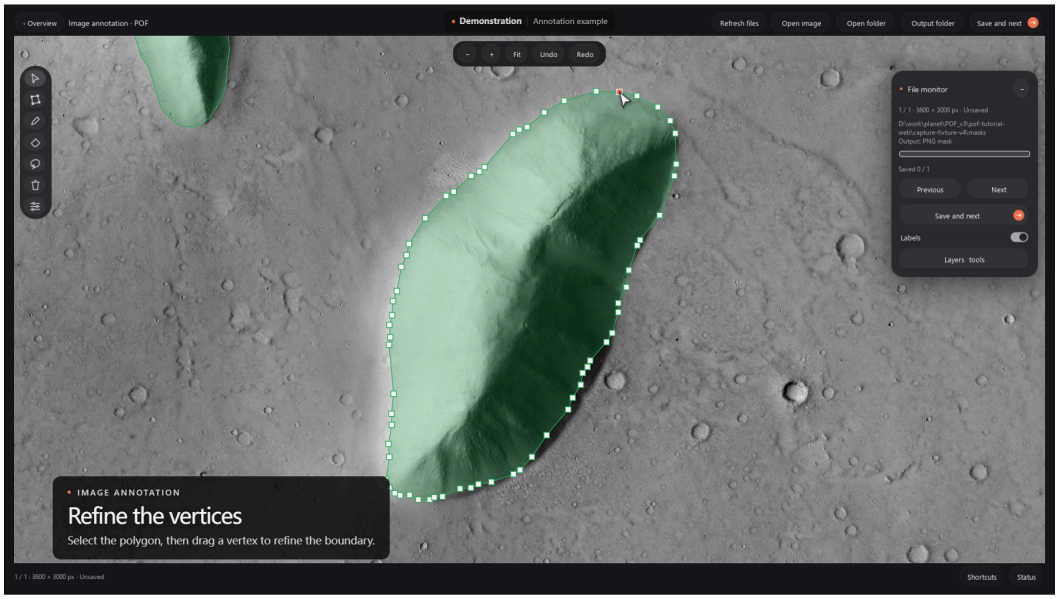
1. Choose **Start annotation**, then **Open image** or **Open folder**. TIFF and common image formats are supported.
2. Choose an **Output folder**. Existing same-name PNG masks or SHP labels can be loaded for editing.
3. Draw along the target boundary. Press **Ctrl+S** to save, or choose **Save and next**.

Select the output format in Settings: PNG mask is the default; SHP is also available. When both formats exist for one image, the current output format takes priority. SHP requires a valid spatial reference; PNG masks retain the image pixel dimensions.

Editing tools

- **Select**: select a polygon, drag its vertices, or delete it.
- **Polygon**: click boundary points and double-click to close. Erase mode removes the enclosed area.
- **Brush / Eraser**: drag to paint or erase. Adjust size with the vertical slider, plus/minus controls or numeric input.
- **Lasso**: draw a freehand outline to add or erase a region.
- **Labels**: toggle visibility without deleting labels. Use **Undo / Redo** to reverse edits.

The file monitor shows position, saved state and file progress. Hover over a tool to see its shortcut; review or change bindings in the shortcuts window.



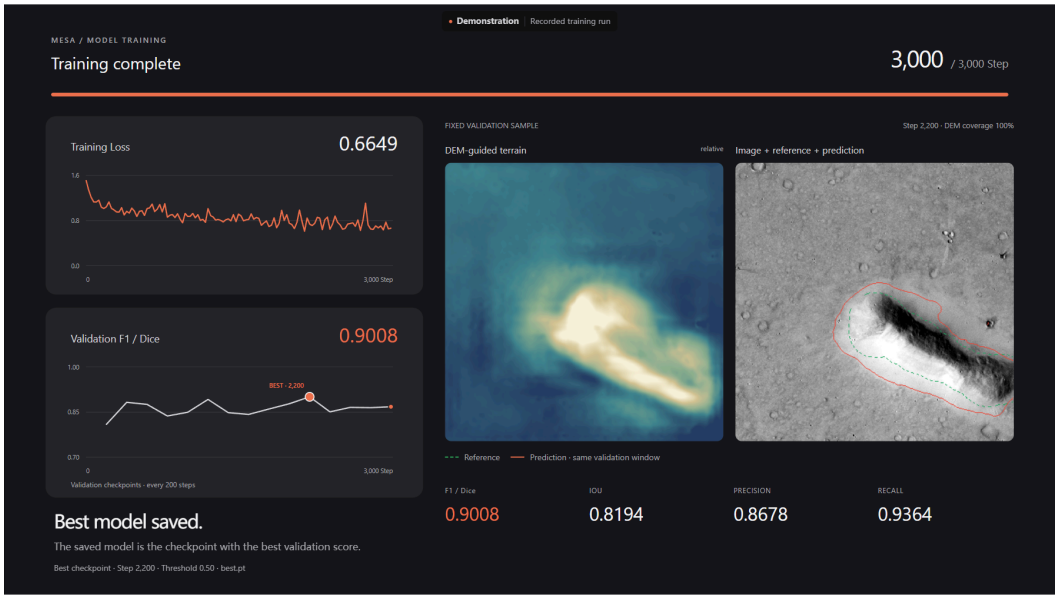
Dark English annotation interface.

3. Training

1. Select image and label folders in the task data page. Run **Check data** and enter **Training** after verifying valid pairs.
2. Select an initialization model and training steps. Adjust Batch, learning rate and other controls in advanced settings. Use built-in **POFBase** when no task model is available.
3. Use **Add DEM** if available, or leave it empty. Confirm the device and settings, then choose **Start training**.
4. Watch Loss, validation Dice/F1 and fixed examples. Review the best model and result details before using it for prediction or active learning.

Loss measures training error. Validation Dice/F1, IoU, Precision and Recall describe segmentation quality; inspect example boundaries as well as numbers. The best model is selected by validation performance and may come from an earlier step.

Use **Safe stop** and wait for completion when interrupting a run. If a model was saved but post-training testing did not finish, use the follow-up test action rather than retraining.



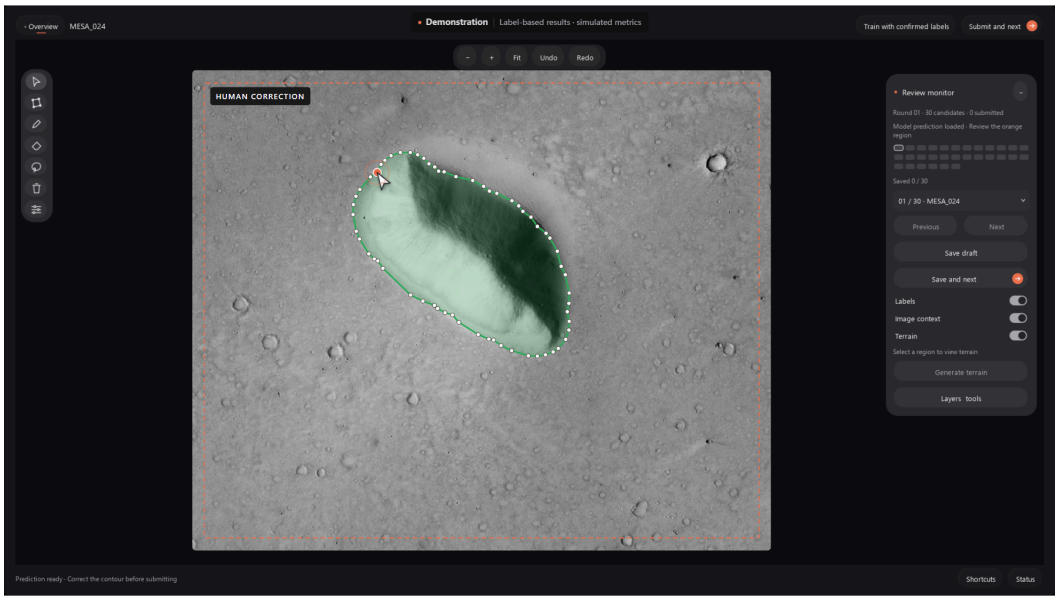
Training status, validation curves and fixed examples.

4. Active learning

Active learning focuses manual review on recommended regions, then updates the model with confirmed labels. It requires an accessible unlabeled image pool.

1. Add an unlabeled image folder to the task, then enter **Active learning**. Choose an existing starting model or use the task model selected automatically.
2. With a trained model, recommend regions and correct its suggestions. With POFBase, complete initial manual annotations before training the first task model.
3. Enter the review canvas. The orange dashed boundary marks the review area. Adjust it and check every target inside. Use **Terrain** when terrain assistance is available.
4. Correct the suggested labels and choose **Submit and next**. You do not need to finish every region: **Train with completed annotations** uses the completed subset.
5. After training, review the round model, metrics and examples in the overview, then start another recommendation round.

On submission, unmarked pixels inside the reviewed area become background; pixels outside remain unknown. Unreviewed regions are not automatically treated as background. Compare rounds under the same validation conditions.



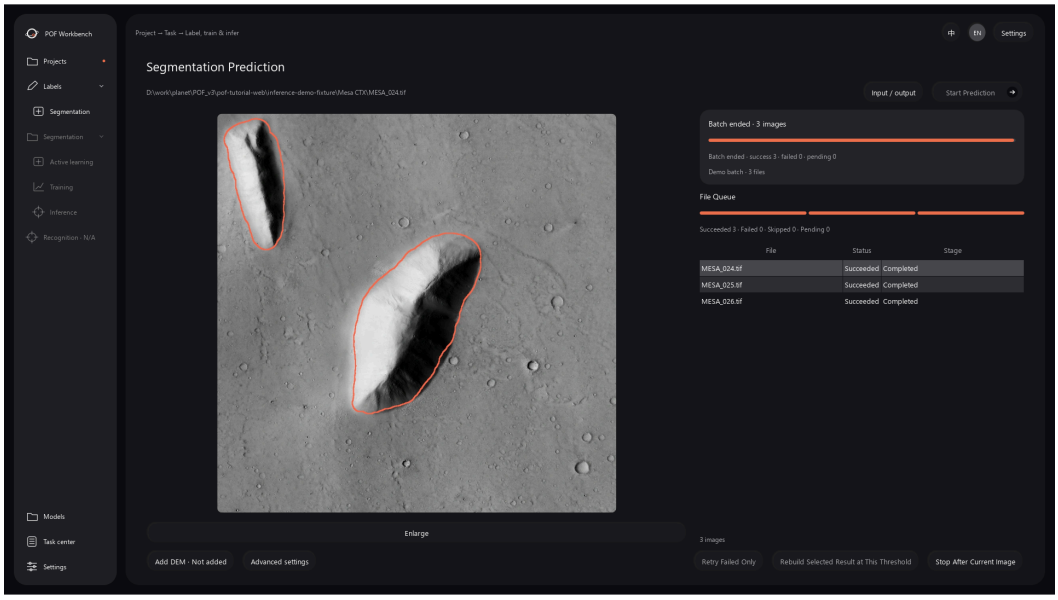
Active learning review canvas. Tutorial gains are illustrative, not performance claims.

5. Batch inference

1. Open **Inference**. In **Input / output**, confirm the complete segmentation model and input/output folders.
2. Add DEM if needed. Choose one output folder per image or grouped folders for mask, Shapefile and other formats.
3. Scan the images, check the queue and choose **Start prediction**.
4. Inspect completed boundaries at a larger scale. To correct results, open the source image and same-name label in the annotation page.

Inference uses the scale windows stored with the model. Target-size filtering removes results without changing model training. PNG masks support further editing; spatial vector outputs can be checked in GIS.

Read failures beside the file or in task records. After checking paths, storage and model availability, use **Retry failed only**. Supported saved results can be regenerated after changing the threshold.



File status and processing stages in the inference queue.

6. Models, DEM and settings

- **Models**: import a complete model, check its source and validation conditions, then use it for annotation, training or inference. Defaults prioritize the selected best model for the task; a starting model can also be chosen manually.
- **DEM**: optional. If empty or unavailable for the current window, supported POF models fall back to monocular relative terrain. Relative terrain is not measured elevation.
- **Settings**: change language and appearance, including system theme. Download GPU runtime and MarsScope only when needed, then check the configured status.
- **Task center**: inspect stages, errors and full records. Leaving a page does not stop a task.

7. Troubleshooting

- **Active learning cannot start:** enter a task and configure an unlabeled image pool. Use Training when you only have paired training data.
- **Labels do not load:** check names, pixel dimensions and the current output-format priority.
- **GPU unavailable or out of memory:** check runtime configuration and reduce Batch. Use CPU to verify the data workflow if needed.
- **Missing file or output failure:** check network access and free disk space. Move deeply nested Windows projects to a shorter local path.
- **Unexpected metrics:** inspect label foreground and fixed validation examples for empty labels, wrong pairs or class-encoding errors.

Report reproduction steps, task records and the complete error message. Back up source images, labels and models. Only clear caches that are confirmed to be rebuildable.